

the data. They comprise the group that can potentially benefit from the analysis of the data.

Physical layer: This layer consists of the physical hardware of the IoT ecosystem. This includes the devices, embedded sensors, networking gear, physical gateways/switches, etc.

Network layer: This layer is mainly responsible for transferring the data generated and collected at the physical layer, to other devices.

Application layer: This layer is mainly intangible as it holds the protocols used for sharing data across heterogeneous devices. It also consists of the interface that helps different devices identify and communicate with each other efficiently.

Remote: The remote allows entities to control and connect to their IoT devices through a dashboard such as an app. Examples of remotes are PCs, smartwatches, connected TVs, tablets, smartphones, etc.

Dashboard: The dashboard is included in the remote, where it allows the entities to control and manage the IoT ecosystem.

What is Big Data?

Big Data refers to the large volume of both structured and unstructured data. Big Data can be mined for insights and information. Data these days runs into exabytes. There are '5 Vs' of Big Data.

Volume: This refers to the amount of data that is generated all over the world. Ninety per cent of the world's data has been generated in the last two years.

Velocity: This is the speed at which the data is generated as well as the speed at which it travels. For example, the New York Stock Exchange creates over 1TB of data daily.

Variety: This refers to the different forms of data generated including structured, unstructured and semi-structured data. Eighty per cent of the world's data is unstructured.

Veracity: This refers to the accuracy and reliability of the data. Uncertainty in data due to inconsistency and incompleteness leads to losses to companies that can add up to millions of dollars.

Value: Value signifies the yield and advantage that the data provides to businesses in the form of insights provided by analysing and mining the massive data.

The intersection of Big Data and IoT

As there are multitudinous sensors and smart devices all over the globe, IoT triggers an inundation of data or Big Data. Only Big Data technologies and frameworks can handle such colossal data volumes that are streaming varied types of information. The more the IoT grows quantitatively, the more Big Data techniques will be required. Within this space, organisations need to shift focus to the rich data, which is easily accessible in real-time. Such data affects the customer base and can generate meaningful conclusions through mining. Data from sensors should be processed to find patterns and insights in real-time to advance business goals. Existing Big Data technologies can effectively harness the incoming sensor

The Internet of things Ecosystem

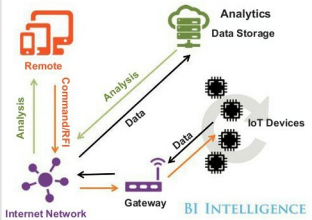


Figure 1: The IoT ecosystem

data, store it and later analyse it efficiently using artificial intelligence. Effectively, for IoT processing, Big Data is the fuel and artificial intelligence is the brain.

Benefits from the intersection of IoT and Big Data

In the present day, over half of all IoT activity is in the fields of transport, manufacturing, user applications, smart cities, etc. IoT will create new business opportunities in the following ways.

New business models: Companies could create value streams for clients, speed time to market and react quickly to client demands.

Real-time information on mission-critical systems: Companies can collect data about products and processes quickly, and improve market agility.

Diversification of revenue streams: Enterprises can monetise more services in addition to the conventional business services.

Global visibility: Enterprises can have better insights into their business, like tracing the path of a component from one extreme of a supply chain to another, which reduces the cost of business in distant localities.

Efficient, intelligent operations: Information from independent endpoints can be accessed by companies to make impromptu decisions on sales, logistics, etc.

Data storage solutions: PaaS

As the continuous streams of machine data from IoT require huge physical storage, organisations are migrating to PaaS (Platform-as-a-Service). This eliminates the need for companies to have their own storage infrastructure, which would need continuous expansion to accommodate the increasing data. PaaS provides easy scalability, compliance, flexibility and a sophisticated architecture that is specially customised to handle IoT data. Moreover, one can opt for private, public or hybrid cloud platforms. Private platforms cater to only a single